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Personalized treatment selection in routine care: Integrating machine learning and statistical algorithms to recommend cognitive behavioral or psychodynamic therapy

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CLINICAL & AGENTIC AI TAKEAWAY

A treatment selection algorithm using machine learning and statistical inference may improve treatment outcomes for some outpatients, but not all, and support therapists' clinical decision-making.

ABSTRACT

Objective: This study aims at developing a treatment selection algorithm using a combination of machine learning and statistical inference to recommend patients' optimal treatment based on their pre-treatment characteristics. Methods: A disorder-heterogeneous, naturalistic sample of N = 1,379 outpatients treated with either cognitive behavioral therapy or psychodynamic therapy was analyzed. Based on a combination of random forest and linear regression, differential treatment response was modeled in the training data (n = 966) to indicate each individual's optimal treatment. A separate holdout dataset (n = 413) was used to evaluate personalized recommendations. Results: The difference in outcomes between patients treated with their optimal vs. non-optimal treatment was significant in the training data, but non-significant in the holdout data (b = -0.043, p = .280). However, for the 50% of patients with the largest predicted benefit of receiving their optimal treatment, the average percentage of change on the BSI in the holdout data was 52.6% for their optimal and 38.4% for their non-optimal treatment (p = .017; d = 0.33 [0.06, 0.61]). Conclusion: A treatment selection algorithm based on a combination of ML and statistical inference might improve treatment outcome for some, but not all outpatients and could support therapists' clinical decision-making.

Keywords: Variable selection, Machine Learning, Random Forest, Outcome Prediction, Precision Medicine, Outpatient Psychotherapy

Study Overview & Background

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Methodology & Clinical Focus

Study Classification: theoretical, modeling, or simulation study. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

A treatment selection algorithm using machine learning and statistical inference may improve treatment outcomes for some outpatients, but not all, and support therapists' clinical decision-making.

Extended Repository Abstract

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Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

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